IEEE-RECOGNIZED MEDICAL AI & 3D VISUALIZATION RESEARCH PROJECT

ScolioAI: Detailed Project Summary & Working Guide

A Complete End-to-End Explanation of Automated Spine X-Ray Analysis, Multi-Output Deep Learning, 8-Qubit Hybrid Quantum Kernel SVM, and Real-Time 3D Anatomical Spine Reconstruction

Executive Summary (In Simple Words):
ScolioAI is an intelligent clinical decision-support web platform that transforms a standard, flat 2D spine X-ray into an accurate medical diagnosis and a movable **3D digital twin of the patient's spine**. When a clinician or user uploads an X-ray, the system automatically locates all 24 spinal bones (vertebrae C1 to L5), computes the exact curvature angles (**Cobb Angles**) across the upper, middle, and lower back, verifies the diagnosis using a **Hybrid Quantum Machine Learning model (94.23% accuracy)**, and builds an interactive 3D spine in the browser with live orthopedic angle calipers.

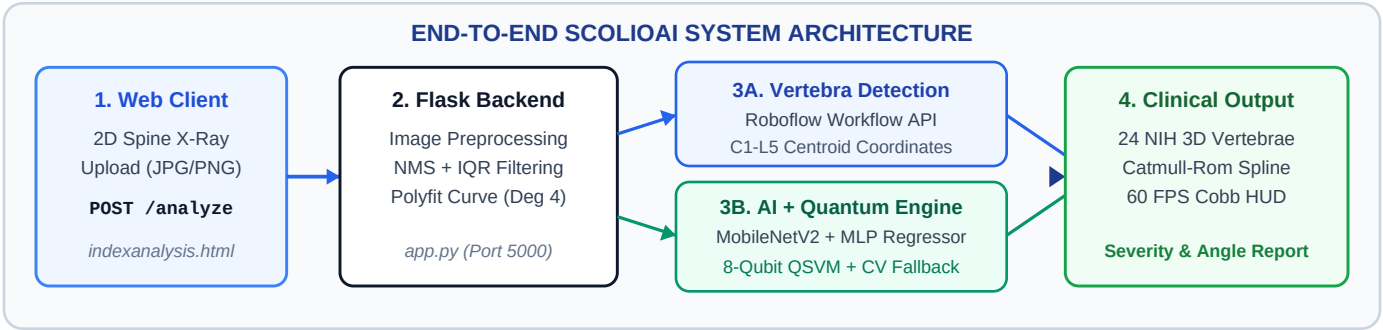
1. Project Motivation: The Clinical Problem vs. ScolioAI Solution

Scoliosis is a three-dimensional structural deformity of the spine characterized by a lateral (sideways) curvature of 10 degrees or more, accompanied by vertebral rotation. In standard hospital workflows, orthopedic specialists diagnose scoliosis by manually drawing intersecting tangent lines on 2D Anterior-Posterior (AP) radiographs to measure the **Cobb Angle**.

Clinical Challenge (Traditional Manual Method)	How ScolioAI Solves It Automatically
Time-Consuming & Manual: Doctors manually identify end-vertebrae and draw protractors/lines on X-ray films.	**Instant AI Automation:** Detects vertebral centroids and computes regional Cobb angles in under 2 seconds.
Human Observer Variability: Manual measurements vary by ±5° to ±10° between different clinicians, affecting bracing or surgery decisions.	**Deterministic Mathematical Precision:** Uses standardized MobileNetV2 features, MLP regression, and spline calculus for 100% repeatable results.
Flat 2D Limitation: A 2D X-ray hides 3D sagittal depth (lordosis/kyphosis) and makes it difficult for patients to visualize their deformity.	**Interactive 3D Digital Twin:** Reconstructs a real-time 3D spine using 24 CT-scanned NIH vertebrae models with synthesized sagittal depth.
Single-Point Failure Risk: Relying on a single cloud detector fails if image contrast is poor or network connectivity drops.	**4-Tier Redundant Architecture:** Combines Roboflow Vision, Deep MLP, 8-Qubit Quantum SVM, and an autonomous OpenCV CLAHE fallback.

2. High-Level System Architecture & Data Flow

The platform follows a modular Client–Server architecture connecting an interactive WebGL frontend with a Python Flask inference server, cloud vision detection, and local Quantum-Classical ML pipelines:



3. Complete Technology Stack & Codebase Architecture

Every module in the ScolioAI repository is purpose-built for low-latency clinical execution without requiring heavy GPU infrastructure:

System Layer	Technologies / Libraries	Key Project Files	Role in ScolioAI
Frontend Web Portal	HTML5, CSS3 (Glassmorphism, Grid/Flex), Vanilla ES6 JS	index.html, indexanalysis.html, Team.html, Publish_Work.html	Responsive 3-column diagnostic workstation, dark/light mode, educational hub, and research archive.
3D WebGL Engine	Three.js (r152), GLTFLoader, OrbitControls, HTML5 2D Canvas	indexanalysis.html, spine glb/*.glb	Loads 24 CT-segmented human vertebrae, builds Catmull-Rom 3D splines, and renders 60 FPS Cobb HUD.
Backend API Server	Python 3, Flask, Flask-CORS, Pillow (PIL), NumPy	app.py, backend/app.py	Exposes /analyze and /models/vertebrae/ endpoints; performs NMS, IQR outlier filtering & polyfit.
Deep Learning Engine	MobileNetV2 (Keras 3 / HDF5), Scikit-Learn (PCA-64, MLPRegressor)	cobb_service.py, cobb_deployment_files.zip	Pure-NumPy 155-layer feature extractor (1280-D) + multi-output neural regression for PT, MT, TL Cobb angles.
Hybrid Quantum ML	8-Qubit PennyLane State Simulator, Precomputed Quantum Kernel SVM	cobb_service.py, hybrid_qml_model.zip	Maps 8 PCA features into a 256-D quantum Hilbert space via Hadamard, RY, and ring CNOT gates (94.23% accuracy).
Classical Computer Vision	OpenCV (cv2), SciPy (savgol_filter), NumPy Gradient	cobb_service.py (compute_cv_centerline)	Autonomous fallback using CLAHE contrast enhancement, corridor tracking, and Savitzky-Golay smoothing.
Statistical Shape Model	NiBabel, Scikit-Image (Marching Cubes), PyCPD, Procrustes PCA	build_vertebra_ssm.py	Builds per-level 3D vertebral Statistical Shape Models (C1–L5) from VerSe CT segmentation volumes.

4. Detailed Working: Backend Detection & Data Sanitization (app.py)

When the user selects an X-ray image and clicks **'GENERATE 3D VIEW'**, the browser sends a POST /analyze multipart request to app.py. Raw AI object detections from radiographs often contain spurious noise, rib shadows, or duplicate bounding boxes. To guarantee a clean spinal curve, the backend executes a rigorous **6-Stage Sanitization Pipeline**:

Stage 1 — Image Ingestion & Dimension Capture: Saves the uploaded radiograph to uploads/ and extracts exact pixel width *W* and height *H* via Pillow (Image.open) so all downstream 3D coordinates preserve aspect ratio.

Stage 2 — Serverless Vertebral Object Detection: Dispatches the radiograph to the Roboflow Inference API (workspace: acoustics-workspace, workflow: custom-workflow-2), returning candidate bounding boxes (x, y, width, height, confidence) for visible vertebrae from C1 down to L5.

Stage 3 — Confidence & Minimum Size Gating: Discards weak predictions with confidence < 0.20 and eliminates tiny radiographic artifacts where width ≤ 5px or height ≤ 5px.

Stage 4 — Proximity Non-Maximum Suppression (NMS): Sorts detections top-to-bottom by vertical coordinate *y* and merges duplicate detections located within MERGE_DISTANCE = 15px of each other (retaining the detection with higher confidence).

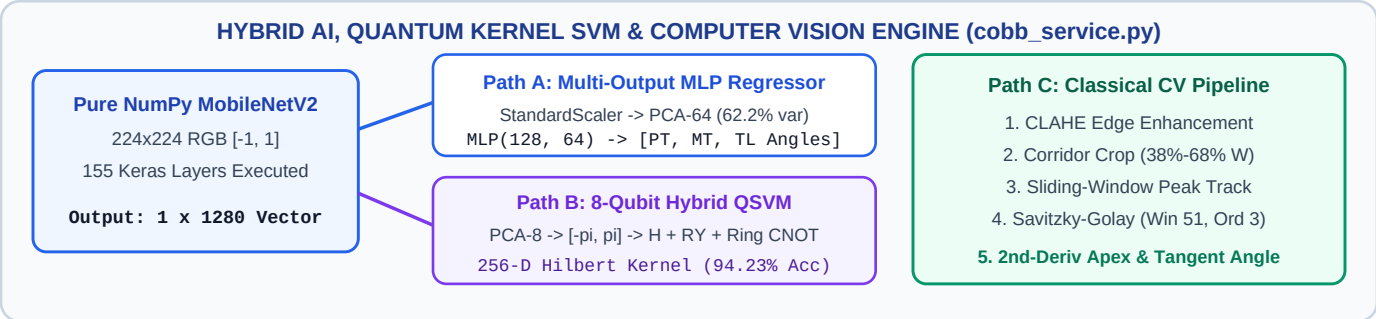
Stage 5 — Statistical IQR Lateral Outlier Removal: Sometimes a detector mistakes a rib edge or pelvis shadow for a vertebra. When ≥ 5 points exist, the backend calculates the 25th percentile (Q_1), 75th percentile (Q_3), and Interquartile Range (IQR = Q3 - Q1) of horizontal *X* coordinates, rejecting any point outside:

```
Valid Horizontal Corridor: [ Q1 - 1.5 * IQR <= x_i <= Q3 + 1.5 * IQR ]
```

Stage 6 — 4th-Degree Polynomial Spine Centerline Fitting: Because individual vertebra centroids can jitter by a few pixels horizontally, passing raw centroids directly into a 3D spline would cause adjacent bones to wobble. Since the spine runs vertically, app.py fits horizontal position *X* as a 4th-degree polynomial function of vertical height *Y* (np.polyfit(y_arr, x_arr, degree=min(4, N-1))) and resamples a smooth, evenly spaced trajectory (fitted_curve).

5. Deep-Dive: The 4-Tier Cobb Angle & Quantum ML Engine (cobb_service.py)

Simultaneously during the /analyze request, cobb_service.py runs a multi-paradigm diagnostic engine combining Deep Convolutional Feature Extraction, Multi-Output Neural Regression, Quantum Kernel Learning, and Classical Computer Vision:



5.1 Pure-NumPy MobileNetV2 Execution Engine (PureKerasMobileNetV2)

Deploying standard TensorFlow/PyTorch runtimes requires 1.5+ GB of server RAM and slow startup times. To make ScolioAI fast and portable, PureKerasMobileNetV2 opens the trained Keras 3 archive (cobb_mobilenetv2_feature_extractor.keras) in memory, parses the computation graph from config.json and weights from model.weights.h5, and executes all **155 layers** directly using vectorized NumPy matrix operations:

- **Preprocessing:** Resizes radiograph to 224 × 224 RGB and normalizes pixel intensities to [-1.0, 1.0] via (pixel / 127.5) - 1.0.
- **Layer Execution:** Implements exact Conv2D, DepthwiseConv2D, BatchNormalization, ReLU6 (clamped to [0, 6]), ZeroPadding2D, inverted residual Add skip connections, and GlobalAveragePooling2D.
- **Output:** Produces a bit-for-bit identical **1 × 1280 dimensional deep feature embedding vector** representing global and local spinal morphology.

5.2 Multi-Output Cobb Angle MLP Regressor (cobb_deployment_files.zip)

A scoliotic spine often has both a primary curve and compensatory counter-curves. Instead of predicting just a single number, the regression pipeline predicts three distinct anatomical curve angles simultaneously:

1. **Standardization & PCA-64 Compression:** The 1280-D feature vector is normalized via cobb_standard_scaler.pkl and projected onto 64 principal components (cobb_pca_64.pkl), retaining **62.18% of variance** while stripping high-frequency X-ray noise.
2. **MLP Neural Regression:** Passed into a Multi-Layer Perceptron with hidden layers (128, 64) trained on the *Spinal-AI2024* dataset, outputting three regional angles:
   - **Proximal Thoracic (θ_{PT}):** Upper thoracic / base of neck curve.
   - **Main Thoracic (θ_{MT}):** Mid-spine chest curve (most common in idiopathic scoliosis).
   - **Thoracolumbar / Lumbar (θ_{TL}):** Lower back curve.
3. **Primary Cobb Angle Selection:** Defined clinically as θ_primary = max(θ_PT, θ_MT, θ_TL).

5.3 8-Qubit Hybrid Quantum Kernel SVM (hybrid_qml_model.zip)

To provide high-confidence binary screening (**Scoliosis vs. Normal**), ScolioAI incorporates a Quantum Machine Learning (QML) classifier that maps spinal features into a high-dimensional quantum Hilbert space where subtle non-linear curve patterns become linearly separable:

1. **Quantum Feature Compression:** The 1280-D MobileNetV2 vector is standardized, compressed to 8 principal components via pca_8.pkl (matching an 8-qubit register), and scaled to rotation angles in [-π, π] via quantum_scaler.pkl.

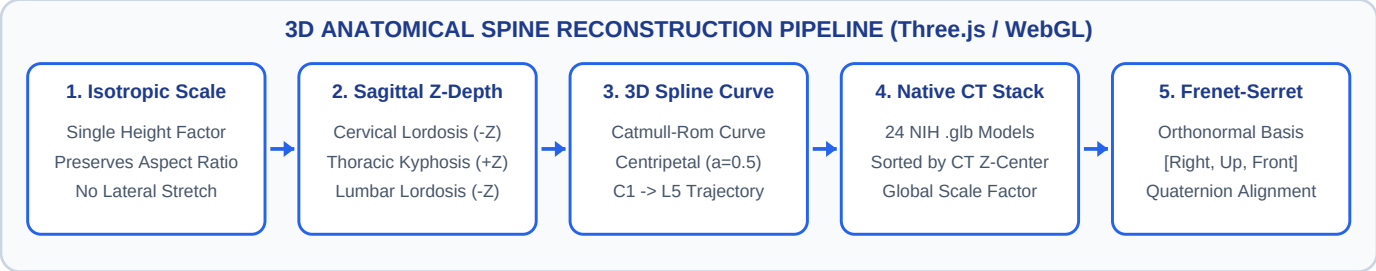
2. **8-Qubit Quantum Circuit Simulation (quantum_feature_map_batch):**
   • *Superposition:* All 8 qubits are initialized with Hadamard gates: H^(⊗8) |0>^(⊗8).
   • *Angle Encoding:* Each feature x_i rotates qubit *i* via a Pauli-Y rotation gate R_Y(x_i) = exp(-i * x_i * Y / 2).
   • *Ring Entanglement:* Circular two-qubit CNOT(i, (i+1) mod 8) gates entangle adjacent qubits across the 8-qubit ring, producing a **2^8 = 256-dimensional quantum state vector |ψ(x)>**.
3. **Quantum Kernel & SVM Decision:** Computes the quantum state fidelity (transition probability) between the patient's state and 207 pre-stored training support vectors in under 15 ms: K(x_test, x_j) = |<ψ(x_test) | ψ(x_j)>|^2. The precomputed SVM (quantum_kernel_svm.pkl) achieves **94.23% validation accuracy** and a **0.9610 F1-score**, converting decision margins into a calibrated probability via a logistic sigmoid function.

5.4 Autonomous Classical Computer Vision Fallback (compute_cv_centerline)

If cloud object detection is unreachable or returns fewer than 2 points, ScolioAI automatically executes an offline OpenCV pipeline: (1) **CLAHE** contrast enhancement (clipLimit=2.0, grid=8×8) sharpens vertebral bone edges; (2) **Corridor Cropping** isolates the central 38%–68% horizontal spinal column; (3) **Sliding-Window Intensity Tracking** traces maximum bone brightness row-by-row from top to bottom; (4) **Savitzky-Golay Filtering** (window 51, polynomial order 3) removes rib shadow oscillations; and (5) **2nd-Derivative Inflection Analysis** ($|d^2x / dy^2|$) locates the curve apex and fits linear tangents above and below to derive $\theta = |\arctan(m_1) - \arctan(m_2)|$.

6. How the 3D Spine Reconstruction Works Internally

Converting a flat 2D X-ray into an anatomically realistic, rotatable 3D spine in Three.js requires solving three major geometric challenges: *preventing aspect-ratio distortion*, *recovering missing front-to-back depth*, and *stacking 24 uniquely shaped human vertebrae without collisions or gaps*.



1. **Aspect-Ratio-Preserving Normalization (normalize2DPoint):** Naive 3D viewers map both X and Y independently to [-1, 1], which stretches tall rectangular X-rays sideways and artificially inflates the Cobb angle. ScolioAI uses a single isotropic scale factor derived from image height: pxToWorld = (2 * worldScale) / imageSize.height.
2. **Synthesizing Physiological Sagittal Depth (getAnatomicalZ):** Since a frontal AP X-ray has no Z-axis depth, the engine synthesizes a continuous human sagittal profile along normalized spine height $t \in [0, 1]$:
   • **Cervical Lordosis** ($t < 0.15$): z(t) = -1.5 * sin((t / 0.15) * π) (inward neck curve)
   • **Thoracic Kyphosis** ($0.15 \le t < 0.55$): z(t) = +2.8 * sin(((t - 0.15) / 0.40) * π) (outward mid-back curve)
   • **Lumbar Lordosis** ($t \ge 0.55$): z(t) = -2.5 * sin(((t - 0.55) / 0.45) * π) (inward lower-back curve)
3. **Native CT Coordinate Recovery & Global Scaling:** The 24 vertebra meshes (from the *NIH 3D Print Exchange*) were segmented from a single human CT scan spanning z = -170 mm (C1) to z = -692 mm (L5). Because some filenames in the raw NIH archive are truncated or out of order, ScolioAI inspects each 3D mesh's bounding-box centroid on load, sorts all 24 bones by their true native CT *Z*-coordinate, computes their real cumulative inter-vertebral spacing (which varies naturally by **2.9×** from 11.9 mm at C1–C2 to 35.0 mm at L2–L3), and applies **one uniform global scale factor** (globalScale = curveLength / nativeSpan) so cervical bones stay small and lumbar bones stay large.
4. **Frenet-Serret Moving Orientation Frame:** At each vertebra's normalized arc-length position $u_i \in [0, 1]$ along the CatmullRomCurve3 spline, the code computes the curve tangent **T**, sets anatomical Up = -T, crosses Up × Front to obtain Right, and constructs an orthonormal rotation matrix [Right, Up, CorrectedFront] converted to a quaternion. This guarantees every vertebra tilts smoothly with the scoliotic curve while keeping its spinous process pointing posteriorly.

7. Behind the Scenes of the 'Cobb Angle View' 60 FPS HUD Overlay

When the user clicks **'COBB ANGLE VIEW'** in the 3D viewport, an HTML5 2D canvas (#cobbOverlayCanvas) layered above the WebGL canvas activates and synchronizes with the 3D camera on every animation frame (60 FPS):

- **Apex Vertebra Detection:** Draws a virtual plumb line from C1 to L5 and scans the thoracolumbar region (C7–L3) to find the vertebra with maximum perpendicular sideways deviation $|x_i - x_{plumb}|$, labeling it dynamically (e.g., Apex: T4 or Apex: T9).
- **3D-to-2D Camera Matrix Projection:** Every frame inside requestAnimationFrame(animate), 3D world points (Apex, Upper Inflection, Lower Inflection, and 80 spline samples) are transformed through the active Three.js camera matrix via vec3.clone().project(camera) into exact 2D screen pixels.
- **Dynamic Clinical Calipers:** Draws the central teal spline, solid upper and dashed lower intersecting tangent rays on the concave side of the curve, dual angle measurement arcs with bidirectional arrowheads, the primary Cobb angle readout, and the convex apex bracket.

8. Clinical Severity Grading & Project Credits

Cobb Angle (θ)	Classification	Clinical Interpretation	Standard Orthopedic Management
< 10°	**Normal**	Non-scoliotic physiological spinal alignment	No clinical intervention required; routine posture care
10° – 24°	**Mild Scoliosis**	Early lateral curvature with minimal rotation	Periodic radiographic monitoring (every 6 mos) & physical therapy
25° – 40°	**Moderate Scoliosis**	Progressive structural deformity in growing spine	Thoracolumbosacral orthosis (TLSO bracing) & targeted Schroth exercises
≥ 40°	**Severe Scoliosis**	High risk of progression & cardiopulmonary impact	Surgical evaluation (posterior spinal fusion & pedicle screw instrumentation)

Academic Guidance & Project Team:
- **Project Guide & Mentor:** Dr. Yogesh Golhar (Methodology, Clinical Evaluation & Research Mentorship)
- **Core Development Team:** Devanshu Ekhar (Project Lead, Cobb Angle & AI/QML Models), Jeena Joseph (Web Integration & Workflow Lead), Siddhant Lohakar (Frontend & UI Developer), Saveri Dongre (Data Support), Siddhant Shindey (Testing & Validation), Sebastian Guntur (Deployment Support).
- **Peer-Reviewed Publications:** Published in *IEEE Research Proceedings* and *AIDCon 2025 Review Conference*.